Tableau Test: Thought Process

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The Tableau test was the portion of the assessment I spent the least time on. The GIS task took much longer than I anticipated, so Claude Opus 5.5 helped me extensively with this portion. Opus organized the data for me, and I checked the work and tried improving the aesthetics and readability. I did not spend as much time getting familiar with the data as I did for the Python and GIS assessments.

The construction permit layer has one row per work location, so I kept one row per permit (`TrackingNumber`). As in the Python test, I filtered with SQL on DDOT's server. I kept new permits (`RenewalDescription = 'New'`) that were applied for on or after October 1, 2025, issued in FY26 (October 1, 2025 through September 30, 2026), and approved, which left 2,760 permits.

I measured business days from the application date to the last approval date, leaving out weekends and the 12 DC legal holidays in FY26. I used approval rather than issuance because issuance also waits on the applicant paying fees. The layer serves UTC clock times labeled as Eastern, which can move a date across midnight, so I corrected them. After the correction, DDOT's own `DaysToApprove` field matches my dates 99.8% of the time, up from 92.9%. For permit type, I used the category at the start of each work type in `TypeDetailNames`, or "Multiple categories" when a permit has more than one.

Permits took 23.4 business days to approve on average (median 14). The clearest difference is between emergency repairs and planned work. Emergency repairs have an EWR number and are almost all approved within a business day, while planned work averages 33.8. Excavation looks fastest only because it holds most of the emergency repairs. Planned work averages 22.9 business days on first submission and 49.1 when sent back to revise and resubmit. I did spend a few minutes giving the dashboard a DC theme (deep red and white from DC's flag, with charcoal text), only because the default colors were horrible.

With more time, I would have presented the data more clearly and added more worksheets: planned work as the headline, with emergency repairs reported separately; the distribution of business days, not only averages; the time spent in each stage of review; a monthly trend; and where planned work is slowest, by applicant and location. I would also confirm the time stamp correction and the meaning of EWR with DDOT, and look into why permits are sent back.